CS5494 Topics in Generative AI

Assignment

Due: Sunday, 8 November (Week 10)

1. Task description

1.1 You are required to accomplish a Fully Visible Sigmoid Belief Network (e.g., logistic regression) for image generation on the binarized MNIST data.

1.2 You are required to accomplish a neural based autoregressive model (e.g., one or multiple layers) for image generation on the binarized MNIST data.

You may need to first create binarized images yourself (based on MNIST). The generated images do not have to be strictly 28x28 in size, you can choose an appropriate dimension based on your model design. The number of training images is also up to you to decide but ensure there are enough samples to train your autoregressive model, and you need to demonstrate your model's generation capability (e.g., at least 10 images are recommended for submission).

1.3 You are required to implement and train a GAN (you can choose your own encoder and decoder architecture) for handwritten digit generation on the original MNIST dataset (not the binarized version). For the training data, you may choose to use either a subset of the training set or the entire training set, specify your choice (and the size of the subset, if applicable) in your report. After training, the generator must be able to generate MNIST-like digit images, and you need to submit some generated images along with a brief report on training stability (e.g., whether mode collapse occurred).

2. Instructions to hand in the assignment

2.1 Go to Canvas and upload the report. The report should include each member's details (student ID and name). If you work as a group (max 3 students), only one student needs to submit the report which must be named as student ID numbers of all group members separated by underscores. E.g. "xxxxxxxx_xxxxxxxx_xxxxxxxx.**pdf**"

2.2 The report must include **a link of your code and data** (e.g., a shared colab, and google cloud folder for your data, so we can easily run it on colab). Clearly provide instructions on how to run your code in the appendix of the report or include a readme.txt in your shared folder. Your code must contain sufficiently clear comments explaining key components.

Don't update the code/data any more after the submission. If the latest modified time of the shared folder is signiftanly late after the submisison deadline, the whole submission will be taken as a late submission.

2.3 The report must clearly show (i) details of your modules, (ii) some generated results from your model, (iii) training process (e.g., loss curve, generation progress showcase such as saving sample images generated by your model at regular training intervals. Arrange these samples in a grid, e.g., 5 rows × 4 columns, with each row representing a training stage to show how the generation quality improves), (iv) run-time (including training time and inference time).
2.4 There is no special format to follow for the report but please make it as clear as possible.

Late submission:

Suppose you hand in work after the deadline:

If you have not been granted special consideration or arrangements

- A penalty of 5% of the maximum marks will be taken per day (or part) late. After ten days, you will be awarded a mark of zero.
- e.g. If an assignment is worth 40% of the final mark and you are one hour late submitting, then the maximum marks possible would be 38%.
- e.g. If an assignment is worth 40% of the final mark and you are 28 hours late submitting, then the maximum marks possible marks would be 36%.
- Submit early; you can resubmit if there is time before the deadline.

3. Marking scheme

Category	Criterion
Report [100]	Introduction [5] - Brief introduction about every task
	Methods [45] - Pre-processing on data (if any) - The principle of different modules
	Experiments and results (with **Figures or Tables**) [20] - Extensive analysis, including training data, hyperparameter analysis, ablation studies.
	Code [20] - Code runs within a feasible time [10] - Well organized, commented and documented [10]
	Discussion and conclusion [5] - Meaningful conclusion and reflection
	Other [5] - At the discretion of the marker: for impressing the marker, excelling expectation, etc. Examples include fast code, using LATEX, etc.
Code Penalties [-]	Badly written code: [-20] Not including instructions on how to run your code: [-20] Late submission